

Supplemental Texts (DRAFT)

Pooled-sample genotyping of the ZEAL BC₂S₃ population (zealgt)

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1 The crossing scheme and the donor haplotype

Each ZEAL family descends from one teosinte plant crossed to B73. The F_1 receives a single haploid gamete from the teosinte parent, which we call the donor haplotype H_d of donor d . That gamete is a recombinant of the teosinte parent's two haplotypes and exists only in that F_1 ; it is one sequence, with one allele per site, and needs no phasing. The F_1 is $H_d/B73$, so every teosinte base pair in its BC_1 progeny and in all of its BC_2S_3 descendants is a piece of H_d . The heterozygosity of the teosinte parent does not enter any model below.

Single-locus expectation. Let h_t be the probability that a plant of generation t is heterozygous $H_d/B73$ at a locus, and a_t that it is homozygous H_d/H_d . Backcrossing to B73 halves h and creates no homozygotes; selfing halves h and moves the lost half equally into the two homozygous classes:

$$\text{backcross: } h_{t+1} = \frac{1}{2}h_t, \quad a_{t+1} = 0; \quad \text{self: } h_{t+1} = \frac{1}{2}h_t, \quad a_{t+1} = a_t + \frac{1}{4}h_t. \quad (1)$$

From $h_{F_1} = 1$: $h_{BC_1} = \frac{1}{2}$, $h_{BC_2} = \frac{1}{4}$, and after s selfings

$$h_{BC_2S_s} = \frac{1}{4}2^{-s}, \quad a_{BC_2S_s} = \frac{1}{2}\left(\frac{1}{4} - \frac{1}{4}2^{-s}\right). \quad (2)$$

For BC_2S_3 : $h = 1/32 = 3.1\%$, $a = 7/64 = 10.9\%$, $B73 = 55/64 = 85.9\%$, and the donor allele frequency $a + h/2 = 1/8$ is fixed at 12.5% by the two backcrosses, whatever the number of selfings. For BC_2S_2 : $h = 1/16$, $a = 3/32$, $B73 = 27/32$. These are the expectations implemented by `single_locus_p0(n.bc, n.self)` (zealHMM, via `nilHMM::breeding_prior`).

2 Pooled samples: BC_1 pools and BC_2S_3 bulks

Every sequenced library in ZEAL is a pool of six plants: each BC_1 sample is six BC_1 plants of one donor, and each BC_2S_3 skim is a six-plant field-plot bulk of one line. A pool of six diploid plants holds twelve chromosome copies at a locus. Whether those twelve copies can be treated as independent draws depends on where each plant's two gametes come from.

2.1 BC_1 pools: six fixed B73 gametes and six F_1 gametes

A BC_1 plant is a B73 gamete times an F_1 gamete. At a site where H_d carries the ALT allele, the B73 gamete is always REF, and the F_1 gamete carries H_d 's allele with probability $\frac{1}{2}$, independently across plants. The number of carrier plants in pool i is therefore

$$j_i \sim \text{Binom}(6, \frac{1}{2}), \quad f_i = \frac{j_i}{12} \in \{0, \frac{1}{12}, \dots, \frac{6}{12}\}, \quad (3)$$

with $E[f_i] = \frac{1}{4}$, $SD[f_i] = \sqrt{6 \cdot \frac{1}{4} / 12} = 0.102$, and $f_i \leq \frac{1}{2}$ always. Treating the twelve copies as independent haplotypes, each ALT with probability $\frac{1}{4}$ ($\text{Binom}(12, \frac{1}{4})$), would keep the mean but inflate the variance ($12 \cdot \frac{1}{4} \cdot \frac{3}{4} = 2.25$ vs. 1.5 haplotypes) and allow $f > \frac{1}{2}$, which the cross cannot produce. The plant level is what matters here, because the two gametes of a plant have different sources. This is the model of step 4 of discovery (Text 4).

Presence. A segment of H_d is absent from all $6k$ F_1 gametes of a donor's k samples with probability 2^{-6k} , so it is present in the pooled DNA with probability $1 - 2^{-6k}$ (98.4% for one sample, $> 99.99\%$ for three). Completeness of H_d is limited by detection (Text 3), not presence. Along the chromosome j_i is constant between the crossovers of the six F_1 gametes, so neighbouring sites share j_i ; the per-site model of Eq. (3) ignores that correlation.

2.2 BC₂S₃ bulks: sibs of one selfed parent

The six plants of a BC₂S₃ plot are sibs from selfing one BC₂S₂ parent. At a locus the parent is B73/B73, H_d/H_d , or heterozygous, with the BC₂S₂ frequencies of Text 1. If it is homozygous the bulk’s donor fraction is fixed at 0 or 1. If it is heterozygous, every sib gamete carries H_d with probability $\frac{1}{2}$ independently, so each sib’s dosage is $\text{Binom}(2, \frac{1}{2})$ and the bulk’s donor-haplotype count is a sum of six of them:

$$K \sim \text{Binom}(12, \frac{1}{2}), \quad g = \frac{K}{12} \in \{0, \dots, 1\}, \quad E[g] = \frac{1}{2}, \quad \text{SD}[g] = 0.144. \quad (4)$$

Here, unlike the BC₁ pool, grouping the twelve gametes into plants does not change the distribution at a single locus: both gametes of a sib come from the same parent. The marginal donor fraction of a bulk is the mixture

$$P(g) = \frac{27}{32} \delta_0 + \frac{3}{32} \delta_1 + \frac{1}{16} \text{Binom}(12g; 12, \frac{1}{2}), \quad E[g] = \frac{1}{8}, \quad (5)$$

which keeps the 12.5% donor content of Text 1. The bulk’s parent is BC₂S₂. The chr10 genotype frequencies of the two full-coverage mexicana families (84 lines) are closer to the BC₂S₂ expectation, with an excess of HET and a deficit of introgression against it (PLAN_pipeline §4 #12). Within a segment where the parent is heterozygous, K is constant between the breakpoints of the twelve gametes, so a segment’s read fraction is one draw of $K/12$ (e.g. 3/12 or 10/12), not $\frac{1}{2}$. A single-plant three-state model with emission means $\{\epsilon, \frac{1}{2}, 1 - \epsilon\}$ mis-specifies such segments; the alternative is a state space on the parent’s genotype (a BC₂S₂ prior) with Eq. (4) as the heterozygous emission. Which fits the ZEAL bulks better is an empirical question (Text 6).

2.3 Unequal DNA per plant

If plant p contributes a DNA share w_p ($\sum_p w_p = 1$), a read comes from plant p with probability w_p , and both haplotypes of that plant share its weight. The expected ALT fraction becomes $\sum_p w_p x_p / 2$ ($x_p \in \{0, 1, 2\}$ the plant’s dosage) instead of the count-over-12 of Eqs. (3)–(4). With $w_p = \frac{1}{6}$ the equal-pooling models are exact; unequal w_p spreads the fractions off the 1/12 lattice and adds overdispersion shared by the two haplotypes of a plant. **[TO CHECK: measure pooling evenness from the sharpness of the $j/12$ lattice in deep BC₁ samples]**

3 Reads at low coverage

With mean coverage λ per sample, the reads at a site are $\text{Pois}(\lambda)$ (Lander–Waterman). At the BC₂S₃ skim depths:

λ	0 reads	1 read	2 reads	≥ 3 reads
0.25×	0.78	0.19	0.024	0.002
0.4×	0.67	0.27	0.054	0.008
0.67×	0.51	0.34	0.11	0.03
1.2×	0.30	0.36	0.22	0.12

At 0.4× a covered site almost always has one read, which samples one of the bulk’s twelve copies; no single site separates heterozygous from homozygous, and dosage can only come from aggregating reads along a segment (the ancestry HMM of Text 6). The ALT reads at a site thin the Poisson: if the sample’s ALT fraction is f , the ALT reads are $\text{Pois}(\lambda f)$ and the probability of seeing none is $e^{-\lambda f}$ (used in Eq. (8)).

Duplicates. Every model here treats reads as independent draws from the pool. A PCR duplicate is the same molecule counted twice, so it inflates both n and the ALT count a without adding information, and the likelihood ratios of Text 4, which are sums over reads, overstate the evidence by roughly the duplication factor. At skim depth this is negligible; in the BC₁ samples (5–25× each), where one carrier plant contributes about $\lambda/12$ ALT reads, it can push weak sites over the thresholds. Duplicate removal before variant calling is standard in pooled-sequencing pipelines (Huang et al., 2015) and in GATK best practices (Van der Auwera et al., 2013); the CRISP paper does not discuss it (Bansal, 2010).

4 Per-donor variant discovery

Candidate sites come from CRISP (Bansal, 2010) run on the donor’s BC₁ samples plus a witness pool (the donor’s BC₂S₃ lines merged). Our post-filter (step 4) scores each candidate per donor.

4.1 Pooled likelihood ratio

For pool i of donor d with n_i reads, a_i of them ALT, and a site error rate ϵ ,

$$L_1^{(i)} = \sum_{j=0}^6 \binom{6}{j} 2^{-6} \text{Binom}(a_i; n_i, p_j), \quad p_j = \frac{j}{12}(1 - \epsilon) + \left(1 - \frac{j}{12}\right)\epsilon, \quad (6)$$

$$L_0^{(i)} = \text{Binom}(a_i; n_i, \epsilon), \quad \text{LLR}_d = \sum_{i \in d} \left(\log L_1^{(i)} - \log L_0^{(i)} \right). \quad (7)$$

j_i is marginalised per pool because it is not observed (Eq. (3)); $j_i = 0$ is allowed, so a pool with no ALT reads is not evidence against H_d ’s allele. With prior π the posterior is $\sigma(\text{LLR}_d + \log \frac{\pi}{1-\pi})$; at $\pi = \frac{1}{2}$ the tier thresholds $\text{LLR} \geq 6.9, 4.6, 2.2$ are posteriors $\geq 0.999, 0.99, 0.9$. A site is REF (tier ref) when $\text{LLR} \leq -4$ with pooled depth ≥ 12 .

Site error rate. ϵ is estimated per site from the pools of donors that are confidently non-carriers there ($\text{LLR} < -3$ at the starting value), including two B73 pools counted at the candidate sites, when those hold ≥ 20 reads; otherwise a default is used.

Flags. `af_gt_half`: the donor’s pooled ALT fraction is significantly above $\frac{1}{2}$ (binomial test), impossible under Eq. (3) and the signature of paralogs or mis-mapping. `hidepth`: summed depth $> 2 \times$ the median, the signature of copy-number gain. Tier A additionally requires ALT reads in ≥ 2 BC₁ samples.

4.2 Witness veto

The veto keeps a candidate only if the witness pool shows ≥ 1 ALT read. For a real H_d allele, a line of coverage λ shows no ALT read with probability $E[e^{-\lambda g}]$ over the bulk fraction g of Eq. (5); using $E[e^{-\lambda K/12}] = \left(\frac{1+e^{-\lambda/12}}{2}\right)^{12}$,

$$q(\lambda) = \frac{27}{32} + \frac{3}{32}e^{-\lambda} + \frac{1}{16}\left(\frac{1+e^{-\lambda/12}}{2}\right)^{12}, \quad P(\text{real allele vetoed}) \approx q(\lambda)^L \quad (8)$$

for L lines, treating lines as independent at the locus (they are not: sibling lines share BC₁ and BC₂ ancestry, which makes the true loss larger). $q(0.4) = 0.958$ and $q(1.2) = 0.907$, so the veto

discards about 65% of real alleles with ten lines at $0.4\times$ (0.958^{10}), 19% with 38 lines, and 25% with fourteen lines at $1.2\times$. This is consistent in direction with the pilot, where a 10-line witness kept 20% of CRISP records and a 38-line witness 51% (chr10; records include artifacts, so the numbers are not directly comparable). [TO CHECK: compare Eq. (8) with the QC-set witness]

5 The union of informative sites and gap filling

5.1 Genotype as ancestry times donor allele

For a line of donor d at site s , with ancestry dosage $x \in \{0, 1, 2\}$ (copies of H_d 's segment) and donor allele $D_{ds} \in \{\text{REF}, \text{ALT}\}$, the genotype is fixed:

$$G_s = \begin{cases} x & D_{ds} = \text{ALT} \\ 0 & D_{ds} = \text{REF} \\ \text{unknown} & D_{ds} \text{ missing and } x > 0. \end{cases} \quad (9)$$

A common marker set across donors therefore needs D_{ds} at every site of the union $U = \bigcup_d A_d$ of the donors' tier-A sets, including sites discovered only in other donors. Positions with different ALT alleles in different donors are dropped.

5.2 Gap filling

Own sites ($s \in A_d$) are ALT by discovery and are not re-called. For a gap ($s \in U \setminus A_d$), the donor's BC_1 samples are counted at s (one pileup per sample, all union sites at once) and scored by Eq. (7) without the veto. REF is assigned from the donor's own reads only (tier **ref**). ALT uses a prior raised by the other donors: with k = other donors (in a chosen set, e.g. the same taxon) that carry the allele and m = other donors with a genotype at s ,

$$\pi_{ds} = \frac{w\mu_d + k}{w + m}, \quad P(D_{ds} = \text{ALT} \mid \text{reads}) = \sigma\left(\text{LLR}_{ds} + \log \frac{\pi_{ds}}{1 - \pi_{ds}}\right), \quad (10)$$

a leave-one-out beta-binomial prior with mean μ_d (donor d 's sharing rate: tier A/(A+ref) at its gaps) and weight $w = 2$. ALT is called at posterior ≥ 0.999 ; sites flagged **hidepth** or **af_gt_half** are never promoted. The likelihood ratio models sequencing error but not artifacts; a site that is tier A in another donor, passed that donor's witness and is clean in the B73 pools is independent evidence that the site is real, which is what the prior carries. A mis-mapped paralog shows ALT in all donors, so sharing alone does not separate real alleles from artifacts; the B73 zero class and the flags do.

5.3 Gap filling, step 2: evidence from the lines

Equation (10) leaves a gap missing when the BC_1 reads are neither REF (tier **ref**) nor ALT at posterior ≥ 0.999 . The donor's BC_2S_3 lines add independent evidence, because the ancestry inference (Text 6) tells, for every line i and site s , how many copies of H_d 's segment the line carries, $x_i \in \{0, 1, 2\}$. A read from a donor copy shows D_{ds} ; a read from a B73 copy shows REF whatever D_{ds} is. With n_i reads and a_i ALT reads in line i , and c_s the mappability of the donor copy relative to B73 at s (the share of donor-copy reads that reach s ; the teosinte copy maps less well where it

has diverged),

$$n_i \sim \text{Pois}(k_s \lambda_i [(1 - \frac{x_i}{2}) + \frac{x_i}{2} c_s]), \quad (11)$$

$$a_i \mid n_i, D_{ds} = \text{ALT} \sim \text{Binom}(n_i, \rho_i(1 - \epsilon) + (1 - \rho_i)\epsilon), \quad \rho_i = \frac{\frac{x_i}{2} c_s}{(1 - \frac{x_i}{2}) + \frac{x_i}{2} c_s}, \quad (12)$$

$$a_i \mid n_i, D_{ds} = \text{REF} \sim \text{Binom}(n_i, \epsilon), \quad (13)$$

with λ_i the line’s coverage (from its B73 segments), k_s the site’s B73 mappability (from the B73-ancestry lines at s), and c_s summed out under a per-taxon prior (mexicana: from simulated TIL18 reads against B73 reads, median 0.85, 10.6% of sites below 0.25). Lines with $x_i = 0$ contribute the same term under both hypotheses and cancel. The evidence of the lines, $\text{LLR}_{ds}^{\text{lines}} = \log L(\text{ALT}) - \log L(\text{REF})$, is independent of the BC_1 reads given D_{ds} , so the log-odds add:

$$P(D_{ds} = \text{ALT} \mid \text{BC}_1, \text{lines}) = \sigma \left(\log \frac{\pi_{ds}}{1 - \pi_{ds}} + \text{LLR}_{ds} + \text{LLR}_{ds}^{\text{lines}} \right), \quad (14)$$

and ALT is called at the same cut-off as Eq. (10), posterior ≥ 0.999 . Step 2 never calls REF: one ALT read in a line with $x_i = 2$ is $\sim 1/\epsilon \approx 200$ times more likely under ALT, while ruling ALT out needs many donor-copy reads, and at the sites that step 1 leaves missing there are about 0–1 (≈ 5 donor-segment lines among 40 at $\sim 0.9\times$, the donor copy mapping poorly).

On the simulated QC set (TIL18; 10 lines at $0.8\times$ and $1.2\times$; true and inferred ancestry; truth = the founder’s own reads, see below) Eq. (14) made **no false ALT call at any of the 29,098 truly-REF gap sites**, also with LLR_{ds} set to 0 (false-ALT rate $< 10^{-4}$, 95% upper bound), and recovered 29–46% of the 11,918 truly-ALT gap sites from the lines and prior alone and 93–94% with the BC_1 evidence. On the two mexicana pilot donors (chr10) it filled 2,994 and 2,581 of the 11,819 and 9,079 gaps left by step 1, leaving 10.7% and 7.9% of the union missing.

5.4 Benchmark

On the simulated QC set (founders Gigi and TIL18, five BC_1 pools of $\sim 14\times$, tier A subsampled to the pilot’s sizes; chr10) the first truth was the founder assembly’s AnchorWave SNPs in the lowcopy ranges, with every other position taken as REF. Against it, gap ALT at posterior ≥ 0.999 was truly ALT at the rate of the donors’ own tier-A sites (0.74 and 0.75 vs. 0.68 and 0.74), gap REF was 97–98% correct, and the Bayes rule and the heuristic “tier A or B” rule gave the same precision at this depth. At most of the sites scored as false, however, reads simulated from the same assembly and aligned with minibwa show the non-B73 base at normal depth: the assembly alignment and the read alignment place that sequence differently (or the position is unaligned, or the SNP was lost in the extraction). Because every step of the pipeline works on minibwa-aligned reads, the benchmark now uses as truth what the founder’s own reads show at each site (ALT if $\geq 50\%$ of ≥ 5 reads are ALT, REF if none of ≥ 5 reads is, unknown otherwise). Against it the precision is ≈ 0.97 for own tier-A sites and ≈ 0.99 for gap ALT. This is the right reference for the read-based steps, not a better truth about homology.

6 Ancestry inference, genotype imputation, and their checks

Ancestry inference. Per line, RTIGER (Campos-Martin et al., 2023) on the donor’s own tier-A sites (rigidity 500; RTIGER learns its transition and emission parameters from the data, so no design prior enters). Lines with any chromosome below $2\times$ rigidity covered markers are excluded before inference gives segments with dosage $x \in \{0, 1, 2\}$. These feed the QTL scan (R/qt1 bcsft, Haley–Knott) on a 0.1 cM grid built from the union markers.

Genotype imputation. Per donor, a two-founder PHG graph (B73 and the donor founder built from D_d on U), imputing only that donor’s lines, rasterised at U (Eq. (9)). These feed GWAS.

Checks. (i) Per-line Mb fractions REF/HET/ALT against the single-locus expectation (Text 1; single plant) or the bulk mixture (Eq. (5)), with a Hotelling test on the (HET, ALT) fractions; the expectation for the bulks is that of their BC_2S_2 parent. (ii) B73 controls free of donor segments. (iii) Segment-length distribution against the expectation for bulks of a BC_2S_2 parent. **[TO CHECK: derive the expected segment-length distribution of a bulk from Eq. (4)]**

References

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